

Model Output w/ changing features

All features, PRESENCE_THRESHOLD = 0.5

=====

Results for: Scripps

=====

Total backup species evaluated: 834
Confirmed present (well documented): 224
Confirmed present (underrepresented): 200
Imputed (likely present, unrecorded): 116
False negatives (model missed): 13
True negatives (correctly ignored): 281

Total confirmed + imputed species: 540

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 1859
Confirmed present (well documented): 116
Confirmed present (underrepresented): 248
Imputed (likely present, unrecorded): 116
False negatives (model missed): 99
True negatives (correctly ignored): 1280

Total confirmed + imputed species: 480

=====

Results for: LosMonos

=====

Total backup species evaluated: 85
Confirmed present (well documented): 35
Confirmed present (underrepresented): 28
Imputed (likely present, unrecorded): 7
False negatives (model missed): 4
True negatives (correctly ignored): 11

Total confirmed + imputed species: 70

=====
Results for: MissionBay
=====

Total backup species evaluated: 646
Confirmed present (well documented): 148
Confirmed present (underrepresented): 145
Imputed (likely present, unrecorded): 111
False negatives (model missed): 21
True negatives (correctly ignored): 221

Total confirmed + imputed species: 404

PRESENCE_THRESHOLD = 0.6

=====
Results for: Scripps
=====

Total backup species evaluated: 834
Confirmed present (well documented): 214
Confirmed present (underrepresented): 175
Imputed (likely present, unrecorded): 55
False negatives (model missed): 48
True negatives (correctly ignored): 342

Total confirmed + imputed species: 444

=====
Results for: ElliottChaparral
=====

Total backup species evaluated: 1859
Confirmed present (well documented): 104
Confirmed present (underrepresented): 180
Imputed (likely present, unrecorded): 56
False negatives (model missed): 179
True negatives (correctly ignored): 1340

Total confirmed + imputed species: 340

=====
Results for: LosMonos
=====

Total backup species evaluated: 85

Confirmed present (well documented): 35
Confirmed present (underrepresented): 28
Imputed (likely present, unrecorded): 7
False negatives (model missed): 4
True negatives (correctly ignored): 11

Total confirmed + imputed species: 70

=====
Results for: MissionBay
=====

Total backup species evaluated: 646
Confirmed present (well documented): 137
Confirmed present (underrepresented): 120
Imputed (likely present, unrecorded): 44
False negatives (model missed): 57
True negatives (correctly ignored): 288

Total confirmed + imputed species: 301

PRESENCE_THRESHOLD = 0.6, without lat/lon

=====
Results for: Scripps
=====

Total backup species evaluated: 834
Confirmed present (well documented): 211
Confirmed present (underrepresented): 176
Imputed (likely present, unrecorded): 57
False negatives (model missed): 50
True negatives (correctly ignored): 340

Total confirmed + imputed species: 444

=====
Results for: ElliottChaparral
=====

Total backup species evaluated: 1859
Confirmed present (well documented): 102
Confirmed present (underrepresented): 186
Imputed (likely present, unrecorded): 57
False negatives (model missed): 175

True negatives (correctly ignored): 1339

Total confirmed + imputed species: 345

=====
Results for: LosMonos
=====

Total backup species evaluated: 85

Confirmed present (well documented): 35

Confirmed present (underrepresented): 28

Imputed (likely present, unrecorded): 7

False negatives (model missed): 4

True negatives (correctly ignored): 11

Total confirmed + imputed species: 70

=====
Results for: MissionBay
=====

Total backup species evaluated: 646

Confirmed present (well documented): 127

Confirmed present (underrepresented): 102

Imputed (likely present, unrecorded): 57

False negatives (model missed): 85

True negatives (correctly ignored): 275

Total confirmed + imputed species: 286

MIN_OBS = 5 (instead of 10),
PRESENCE_THRESHOLD = 0.7, no lat/lon

=====
Results for: Scripps
=====

Total backup species evaluated: 1363

Confirmed present (well documented): 246

Confirmed present (underrepresented): 181

Imputed (likely present, unrecorded): 44

False negatives (model missed): 162

True negatives (correctly ignored): 730

Total confirmed + imputed species: 471

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 2744
Confirmed present (well documented): 94
Confirmed present (underrepresented): 151
Imputed (likely present, unrecorded): 31
False negatives (model missed): 273
True negatives (correctly ignored): 2195

Total confirmed + imputed species: 276

=====

Results for: LosMonos

=====

Total backup species evaluated: 201
Confirmed present (well documented): 66
Confirmed present (underrepresented): 67
Imputed (likely present, unrecorded): 19
False negatives (model missed): 9
True negatives (correctly ignored): 40

Total confirmed + imputed species: 152

=====

Results for: MissionBay

=====

Total backup species evaluated: 976
Confirmed present (well documented): 109
Confirmed present (underrepresented): 83
Imputed (likely present, unrecorded): 35
False negatives (model missed): 213
True negatives (correctly ignored): 536

Total confirmed + imputed species: 227

MIN_OBS = 10, PRESENCE =.5, no
“backup_unique_users”

```
=====
Results for: Scripps
=====
Total backup species evaluated:      834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 199
Imputed (likely present, unrecorded): 142
False negatives (model missed):     13
True negatives (correctly ignored):  255
---
Total confirmed + imputed species:   566
```

```
=====
Results for: ElliottChaparral
=====
Total backup species evaluated:      1859
Confirmed present (well documented): 116
Confirmed present (underrepresented): 254
Imputed (likely present, unrecorded): 162
False negatives (model missed):      93
True negatives (correctly ignored):  1234
---
Total confirmed + imputed species:   532
```

```
=====
Results for: LosMonos
=====
Total backup species evaluated:      85
Confirmed present (well documented): 35
Confirmed present (underrepresented): 30
Imputed (likely present, unrecorded): 13
False negatives (model missed):      2
True negatives (correctly ignored):   5
---
Total confirmed + imputed species:   78
```

```
=====
Results for: MissionBay
=====
```

Total backup species evaluated: 646
Confirmed present (well documented): 149
Confirmed present (underrepresented): 139
Imputed (likely present, unrecorded): 179
False negatives (model missed): 26
True negatives (correctly ignored): 153

Total confirmed + imputed species: 467

W/ Reserve-specific PRESENCE, MIN_OBS

=====
Results for: Scripps
=====
Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 200
Imputed (likely present, unrecorded): 142
False negatives (model missed): 12
True negatives (correctly ignored): 255

Total confirmed + imputed species: 567

=====
Results for: ElliottChaparral
=====
Total backup species evaluated: 1859
Confirmed present (well documented): 115
Confirmed present (underrepresented): 254
Imputed (likely present, unrecorded): 165
False negatives (model missed): 94
True negatives (correctly ignored): 1231

Total confirmed + imputed species: 534

=====
Results for: LosMonos
=====
Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72

Imputed (likely present, unrecorded): 43
False negatives (model missed): 2
True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====
Results for: MissionBay
=====
Total backup species evaluated: 646
Confirmed present (well documented): 136
Confirmed present (underrepresented): 119
Imputed (likely present, unrecorded): 94
False negatives (model missed): 59
True negatives (correctly ignored): 238

Total confirmed + imputed species: 349

Testing ElliotChapparal with MIN_OBS 15, THRESHOLD = 0.6

=====
Total backup species evaluated: 1435
Confirmed present (well documented): 105
Confirmed present (underrepresented): 211
Imputed (likely present, unrecorded): 90
False negatives (model missed): 105
True negatives (correctly ignored): 924

Total confirmed + imputed species: 406

Testing ElliotChapparal with MIN_OBS 10, THRESHOLD = 0.6

=====
Results for: ElliottChapparal
=====
Total backup species evaluated: 1859

Confirmed present (well documented): 111
Confirmed present (underrepresented): 211
Imputed (likely present, unrecorded): 88
False negatives (model missed): 141
True negatives (correctly ignored): 1308

Total confirmed + imputed species: 410

NEW BASELINE

=====

Results for: Scripps

=====

Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 200
Imputed (likely present, unrecorded): 142
False negatives (model missed): 12
True negatives (correctly ignored): 255

Total confirmed + imputed species: 567

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 1435
Confirmed present (well documented): 105
Confirmed present (underrepresented): 211
Imputed (likely present, unrecorded): 90
False negatives (model missed): 105
True negatives (correctly ignored): 924

Total confirmed + imputed species: 406

=====

Results for: LosMonos

=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72
Imputed (likely present, unrecorded): 43
False negatives (model missed): 2

True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====
Results for: MissionBay
=====

Total backup species evaluated: 646

Confirmed present (well documented): 136

Confirmed present (underrepresented): 119

Imputed (likely present, unrecorded): 94

False negatives (model missed): 59

True negatives (correctly ignored): 238

Total confirmed + imputed species: 349

Test without 'backup_unique_days',

=====
Results for: Scripps
=====

Total backup species evaluated: 834

Confirmed present (well documented): 225

Confirmed present (underrepresented): 201

Imputed (likely present, unrecorded): 142

False negatives (model missed): 11

True negatives (correctly ignored): 255

Total confirmed + imputed species: 568

=====
Results for: ElliottChaparral
=====

Total backup species evaluated: 1435

Confirmed present (well documented): 107

Confirmed present (underrepresented): 210

Imputed (likely present, unrecorded): 87

False negatives (model missed): 104

True negatives (correctly ignored): 927

Total confirmed + imputed species: 404

=====
Results for: LosMonos
=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72
Imputed (likely present, unrecorded): 43
False negatives (model missed): 2
True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====
Results for: MissionBay
=====

Total backup species evaluated: 646
Confirmed present (well documented): 136
Confirmed present (underrepresented): 116
Imputed (likely present, unrecorded): 96
False negatives (model missed): 62
True negatives (correctly ignored): 236

Total confirmed + imputed species: 348

SAME obs, threshold missing the features below

```
"""
    features not used in this test:
    'lat_std',
    'lon_std',
    'backup_unique_users',
    'avg_soil_ph',
    'backup_unique_days',
    'obs_per_user',
    """
```

(venv) (base) carstenpetersen@Carstens-Air CleanedData % uv run EvaluateTest.py

=====

Results for: Scripps

=====
Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 201
Imputed (likely present, unrecorded): 148
False negatives (model missed): 11
True negatives (correctly ignored): 249

Total confirmed + imputed species: 574

=====
Results for: ElliottChaparral

=====
Total backup species evaluated: 1435
Confirmed present (well documented): 104
Confirmed present (underrepresented): 206
Imputed (likely present, unrecorded): 83
False negatives (model missed): 111
True negatives (correctly ignored): 931

Total confirmed + imputed species: 393

=====
Results for: LosMonos

=====
Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72
Imputed (likely present, unrecorded): 43
False negatives (model missed): 2
True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====
Results for: MissionBay

=====
Total backup species evaluated: 646
Confirmed present (well documented): 129
Confirmed present (underrepresented): 95
Imputed (likely present, unrecorded): 94
False negatives (model missed): 90
True negatives (correctly ignored): 238

Total confirmed + imputed species: 318

“Obs_per_user” back in, same threshold, min_obs. NEW BASELINE

=====

Results for: Scripps

=====

Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 201
Imputed (likely present, unrecorded): 142
False negatives (model missed): 11
True negatives (correctly ignored): 255

Total confirmed + imputed species: 568

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 1435
Confirmed present (well documented): 107
Confirmed present (underrepresented): 210
Imputed (likely present, unrecorded): 87
False negatives (model missed): 104
True negatives (correctly ignored): 927

Total confirmed + imputed species: 404

=====

Results for: LosMonos

=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72
Imputed (likely present, unrecorded): 43
False negatives (model missed): 2
True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====

Results for: MissionBay

=====

Total backup species evaluated: 646
Confirmed present (well documented): 136
Confirmed present (underrepresented): 116
Imputed (likely present, unrecorded): 96
False negatives (model missed): 62
True negatives (correctly ignored): 236

Total confirmed + imputed species: 348

TEST CHANGING MAX_SAMPLES = 0.5 param for PU classifier

=====

Results for: Scripps

=====

Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 205
Imputed (likely present, unrecorded): 180
False negatives (model missed): 7
True negatives (correctly ignored): 217

Total confirmed + imputed species: 610

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 1435
Confirmed present (well documented): 112
Confirmed present (underrepresented): 221
Imputed (likely present, unrecorded): 118
False negatives (model missed): 88
True negatives (correctly ignored): 896

Total confirmed + imputed species: 451

=====

Results for: LosMonos

```
=====
Total backup species evaluated:    201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 73
Imputed (likely present, unrecorded): 48
False negatives (model missed):    1
True negatives (correctly ignored): 11
---
Total confirmed + imputed species: 189
```

```
=====
Results for: MissionBay
=====
Total backup species evaluated:    646
Confirmed present (well documented): 143
Confirmed present (underrepresented): 124
Imputed (likely present, unrecorded): 136
False negatives (model missed):    47
True negatives (correctly ignored): 196
---
Total confirmed + imputed species: 403
```

TEST CHANGING MAX_SAMPLES = 0.4 param for PU classifier

```
=====
Results for: Scripps
=====
Total backup species evaluated:    834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 208
Imputed (likely present, unrecorded): 229
False negatives (model missed):    4
True negatives (correctly ignored): 168
---
Total confirmed + imputed species: 662
```

```
=====
Results for: ElliottChaparral
=====
Total backup species evaluated:    1435
Confirmed present (well documented): 113
```

Confirmed present (underrepresented): 241
Imputed (likely present, unrecorded): 160
False negatives (model missed): 67
True negatives (correctly ignored): 854

Total confirmed + imputed species: 514

=====
Results for: LosMonos
=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 74
Imputed (likely present, unrecorded): 54
False negatives (model missed): 0
True negatives (correctly ignored): 5

Total confirmed + imputed species: 196

=====
Results for: MissionBay
=====

Total backup species evaluated: 646
Confirmed present (well documented): 149
Confirmed present (underrepresented): 134
Imputed (likely present, unrecorded): 179
False negatives (model missed): 31
True negatives (correctly ignored): 153

Total confirmed + imputed species: 462

Max_samples = 0.5 again, increasing threshold by
0.05 on all

=====
Results for: Scripps
=====

Total backup species evaluated: 834
Confirmed present (well documented): 222
Confirmed present (underrepresented): 200
Imputed (likely present, unrecorded): 142

False negatives (model missed): 15
True negatives (correctly ignored): 255

Total confirmed + imputed species: 564

=====
Results for: ElliottChaparral

=====

Total backup species evaluated: 1435
Confirmed present (well documented): 103
Confirmed present (underrepresented): 202
Imputed (likely present, unrecorded): 74
False negatives (model missed): 116
True negatives (correctly ignored): 940

Total confirmed + imputed species: 379

=====
Results for: LosMonos

=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 72
Imputed (likely present, unrecorded): 43
False negatives (model missed): 2
True negatives (correctly ignored): 16

Total confirmed + imputed species: 183

=====
Results for: MissionBay

=====

Total backup species evaluated: 646
Confirmed present (well documented): 134
Confirmed present (underrepresented): 113
Imputed (likely present, unrecorded): 93
False negatives (model missed): 67
True negatives (correctly ignored): 239

Total confirmed + imputed species: 340

Back to baseline

=====

Results for: Scripps

=====

Total backup species evaluated: 834
Confirmed present (well documented): 225
Confirmed present (underrepresented): 205
Imputed (likely present, unrecorded): 180
False negatives (model missed): 7
True negatives (correctly ignored): 217

Total confirmed + imputed species: 610

=====

Results for: ElliottChaparral

=====

Total backup species evaluated: 1435
Confirmed present (well documented): 112
Confirmed present (underrepresented): 221
Imputed (likely present, unrecorded): 118
False negatives (model missed): 88
True negatives (correctly ignored): 896

Total confirmed + imputed species: 451

=====

Results for: LosMonos

=====

Total backup species evaluated: 201
Confirmed present (well documented): 68
Confirmed present (underrepresented): 73
Imputed (likely present, unrecorded): 48
False negatives (model missed): 1
True negatives (correctly ignored): 11

Total confirmed + imputed species: 189

=====

Results for: MissionBay

=====

Total backup species evaluated: 646
Confirmed present (well documented): 143
Confirmed present (underrepresented): 124

Imputed (likely present, unrecorded): 136
False negatives (model missed): 47
True negatives (correctly ignored): 196

Total confirmed + imputed species: 403

Mission Bay max_samples = 0.4

=====

Results for: MissionBay

=====

Total backup species evaluated: 646
Confirmed present (well documented): 149
Confirmed present (underrepresented): 134
Imputed (likely present, unrecorded): 179
False negatives (model missed): 31
True negatives (correctly ignored): 153

Total confirmed + imputed species: 462

Mission Bay max_samples = 0.4, threshold = 0.65

=====

Results for: MissionBay

=====

Total backup species evaluated: 646
Confirmed present (well documented): 143
Confirmed present (underrepresented): 123
Imputed (likely present, unrecorded): 144
False negatives (model missed): 48
True negatives (correctly ignored): 188

Final Preds (my best discretion)